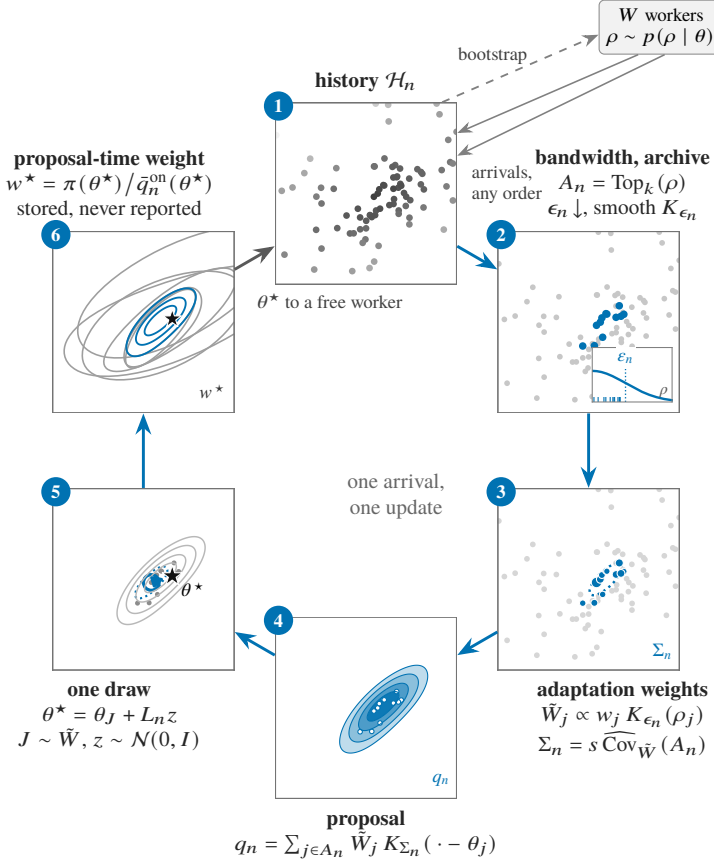




Caption stand-in. One propagator call on the evaluated history (2-D toy: box prior, $k = 12$). Gray shade = ρ_i (light: far); point size = weight; gray contours = stored snapshots q_s ; inset in (2): the kernel $K_{\epsilon_n}(\rho)$ over ρ , ticks = ρ_j of the archive (no acceptance boundary).

(1) $\mathcal{H}_n = \{(\theta_i, \rho_i, \epsilon_i, \tau_i, w_i)\}_{i \leq n}$; every worker's result lands here, in any order. (2) $\epsilon_n = \min(\epsilon_{\text{hist}}, \epsilon_{\text{sched}})$ (Alg. 1 l. 1), A_n (l. 5). (3) $\Sigma_n = L_n L_n^\top$ (l. 5–6). (4)–(5) l. 6.

(6) $\bar{q}_n^{\text{on}} = \frac{1}{1+|\mathcal{S}|} [q_n + \sum_{s \in \mathcal{S}} q_s]$ (l. 7); afterwards q_n is pushed into the ring buffer $\mathcal{S}$ every `amis_interval` calls. $w^\star$ is stored with the particle and never reported: the reported posterior reweights every particle of $\mathcal{H}_n$ (eq. posterior estimator).

Return: $\theta^\star$ goes to a free worker; when $\rho^\star$ arrives, $(\theta^\star, \rho^\star, \epsilon_n, \tau^\star=n, w^\star)$ is appended. Dashed: while $|\mathcal{H}_n| < k$ the call returns a uniform prior draw with $w^\star = 1$ straight to a worker (l. 2–3).